

BASS / HTT Departure-Decomposition Programme

Research-Progress Summary and Planck Raw-Data Analysis Plan

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Abstract

This report summarises the current state of the BASS/HTT FLRW-departure programme and sets out, in detail, the plan for analysing the Planck raw-data (end-to-end simulation) ensemble that closes the one remaining real-data gating measurement (K1). The programme delivers a *diagnostic, provenance-bound* comparator over four anisotropy sectors $g = (\Sigma^2, W^2, \Omega_{\text{tilt}}, \Omega_k)$, a suite of conditional EGS-type (Ehlers–Geren–Sachs) theorems, and three model-independent real-data descriptors (CF4 bulk flow, CF4 curl no-go, low- ℓ CMB morphology). It does *not* establish a Bianchi-family identification, an anisotropic geometry, a native low- ℓ solver validation, a physical-vorticity detection, or a globally significant low- ℓ anomaly. The Planck-raw analysis described in §5 is the matched end-to-end (E2E) null that would upgrade the K1 low- ℓ morphology statistic from a Λ CDM-null look-elsewhere result to an instrument-and-foreground-calibrated one; the full pipeline, three runner modes, and a higher-precision (v2) statistic set are implemented and validated, awaiting only the manual download of the ~ 1 TB simulation set.

1 Programme and framework

The programme tests whether the cosmic microwave background (CMB) and low-redshift peculiar velocities depart from the standard isotropic FLRW cosmology toward an anisotropic (Bianchi + tilt) one. It deliberately works *without* a native low- ℓ anisotropic Boltzmann solver (which does not yet exist in-tree), so it makes no family-identification or geometry-detection claim. Instead, the primary object is the **graded comparator vector**

$$g = (\Sigma^2, W^2, \Omega_{\text{tilt}}, \Omega_k) \in \mathbb{R}^4, \quad x_C = \langle c, g \rangle, \quad c = (+1, -1, +1, +1), \quad (1)$$

where Σ^2 is the shear, W^2 the vorticity, Ω_{tilt} the tilt, and Ω_k the anisotropic-curvature sector; the signed scalar x_C is a *derived* linear summary, so the per-sector content is preserved and the sign-cancellation pathology of a single number is avoided. An equivalent representation casts the comparator as a positive-semidefinite matrix $M = \text{diag}(g) \succeq 0$ with $x_C = \text{tr}(CM)$ (bit-identical), whose admissible set is a convex PSD cone and whose reachable eigen-directions encode identifiability.

Honest claim envelope. All results are diagnostic-only / model-independent / conditional-theorem content. No data-fitting headline exceeds the registry-and-shear-source-validated envelope; the eight Bianchi families beyond {FLRW, I, V, IX} are restricted to axis-aligned mode subsets, and any family/geometry statement remains gated behind the (unbuilt) native solver and its full gate bundle.

2 Theorem programme (progress)

The conditional EGS-type theorem suite spans three axes — mathematics/ statistics, GR/covariant-Boltzmann, and data interpretation — and is gate- and (where stated) Wolfram-verified. The

consolidated ledger currently holds **14 proven rows** (6 closed-form symbolic, 8 gate-verified) and **3 real-data rows**, summarised in Table 1.

Table 1: Consolidated theorem/result ledger (status counts: 6 `proven_symbolic`, 8 `proven_gate`, 1 `measured`, 1 `measured_partial`, 1 `measured_no_go`).

Item	Axis	Statement (abbreviated)
NT-A1	Math/Stat	Closure-conditional quadrupole-filling identity ($\kappa = 4/21$, registered ETM)
NT-A3	Math/Stat	Single-sky sampling dispersion $\sqrt{2/5}$ (one estimator, <i>not</i> a universal floor)
EGS3-A1	Math/Stat	Graded rank-2 identifiability; null = structural $\{W^2\}$ + leading-order $\{\Omega_k\}$
EGS3-A3	Math/Stat	Π as a calibrated e-value (Markov bound; GRF- Λ CDM null)
EGS3-A4	Math/Stat	Rao-Blackwell sufficiency of the reachable-sector statistic
NT2-A1	Math/Stat	Multi- ℓ Fisher floor conditional on the registered shear-response profile
NT2-A2	Math/Stat	Octupole information saturation (tail beyond $\ell \sim 20$ converges)
EGS3-B1	GR/Boltz	Semi-native transfer: floor is a single-mode, finite- k profile
EGS3-B2	GR/Boltz	Volterra depth-memory of the tilt stress; $\equiv$ ODE + Grönwall
NT2-B1	GR/Boltz	Two-sided shear bracket within the registered closure/H3 model
NT2-B3/EGS3-B3	GR/Boltz	Radial-velocity vorticity blindness; transverse channel re-opens
EGS3-B4	GR/Boltz	Covariant bracket constants ($C_{\text{up}}\kappa(1+R) > 1$)
EGS3-PSD	Math/Stat	PSD-cone representation, $x_C = \text{tr}(CM)$, bit-identical
K1	Data	Low- ℓ morphology look-elsewhere p (measured_partial)
K5	Data	CF4 bulk-flow amplitude + conditional coverage (measured)
K6	Data	CF4 WF curl-suppression no-go (measured_no_go)

Sector-null precision (a key refinement). The two unreachable sectors are *not* the same kind of null. W^2 is a genuine, order-independent *structural* null: radial peculiar velocities satisfy $n^a \Omega_{ab} n^b \equiv 0$ exactly and CMB temperature is curl/Weyl-blind at EGS order; its response column is a genuine zero, not collinear with Σ^2 , and it re-opens only via a *different* observable (transverse velocities, CMB B -modes). Ω_k is a leading-EGS-order *no-channel* that re-opens *beyond* leading order (higher-order ISW, lensing, the native transfer) within the same channels.

3 Real-data status

Three real-data descriptors are computed on owned inputs; all are model-independent and carry no family/geometry/anisotropy-evidence claim.

- **K5 (CF4 bulk flow).** A weighted-GLS estimator on the real Cosmicflows-4 group release (Tully et al. 2023, 38,049 groups) gives a bulk-flow amplitude $|\mathbf{B}| \simeq 341 \pm 102 \text{ km s}^{-1}$, the error budget cosmic-variance-dominated; this is consistent with the Λ CDM expectation ($\sim 150\text{--}250 \text{ km s}^{-1}$ at the effective depth), i.e. high but not anomalous. The cosmic-variance-inclusive coverage (0.67 vs 0.19 measurement-only) is computed with geometry-and-error matched Gaussian mock mechanics and is *conditional* on a fixed Λ CDM $\sigma_{\text{cv}} = 150 \text{ km s}^{-1}$ per-component prior; full selection/Malmquist/grouping/correlated-field mocks remain a standing gate.
- **K6 (CF4 curl/vorticity).** The affine velocity-gradient decomposition of the real CF4++ Wiener-filter field returns a vorticity at most 0.55% of the shear at every radius, while an injected solid-body rotation is recovered to machine precision. This is a WF mean-field curl-suppression *structural no-go* (the field is curl-suppressed by the Wiener-filter prior across modes), not a vorticity measurement; a true constrained-realization posterior remains a standing gate.
- **K1 (low- ℓ CMB morphology).** Stacking six registered low- ℓ statistics on the real Planck PR3 maps into a frozen max-scan gives a look-elsewhere-corrected global $p =$

0.097 (SMICA) / 0.121 (Commander) under an isotropic Λ CDM (GRF) null over 2000 realizations. The $\sim 25\%$ method dependence is itself a foreground/cleaning systematic; the two are reported side by side and *not* averaged, and neither is E2E-calibrated. The local single-statistic p are: parity asymmetry 0.021, parity even/odd 0.021, planarity 0.045, $S_{1/2}$ 0.060, Q–O alignment 0.67, axis-to-dipole 0.71. *This is the measurement the Planck-raw plan of §5 upgrades.*

Joint comparator (PR08-006). Assembling the discharges into g on real data gives a data rank of two: Ω_{tilt} is measured (K5 bulk flow) and Σ^2 is partial (K1 look-elsewhere only), while W^2 and Ω_k are fail-closed (K6 no-go; no leading-order channel) — never zeroed. No combined scalar x_C is formed (two sectors are blind), no MIO certificate is used as odds, and no scalar is promoted to a family. The rank-2 count is therefore *one fully measured plus one partial* sector, not two measured sectors.

4 External-audit revision

Two independent external audits (a research-content review, verdict MINOR REVISIONS; and a claim-firewall re-review, verdict MAJOR REVISIONS) were folded in. The substantive items — the W^2 vs Ω_k null distinction, the e-value’s Λ CDM-null idealisation with an add-one finite-null $\hat{\alpha} = (k+1)/(n+1)$, the closure-conditional/registered-response scoping of the Fisher floor and shear bracket — were already realised at the module level and were propagated into all reporting surfaces. The genuine over-claims in the data labels (K5 “release-matched mocks”, K6 “discharged”) were downgraded to the conditional/structural language used above. A permanent claim linter over the report, ledger, and blocker dossier now forbids the flagged phrasings, and an independent re-check script reproduces the bracket constant, the e-value Markov bound, the Fisher-floor descent, the radial-vorticity identity, and the Ω_k genuine-zero demonstration.

5 Planck raw-data analysis plan

5.1 Objective and the standing blocker

The K1 look-elsewhere correction is discharged on the real map under an idealised isotropic Λ CDM (GRF) null. What remains (BLOCKED_MISSING_PR4_E2E_ACCESS) is to replace that null with a **matched end-to-end (E2E) simulation ensemble** carrying real instrument noise, residual foregrounds, and the component-separation transfer, so that the global p is reported with instrument-and-foreground systematics. The ensemble is served only through the Planck Legacy Archive (PLA) interactive query portal (and NERSC-authenticated paths), not as a plain-URL download; *cobaya* was checked and provides the Blackwell–Rao C_ℓ -level low- ℓ likelihood, not the $a_{\ell m}$ map ensemble the morphology statistics require. The acquisition is therefore manual; everything downstream is implemented.

5.2 Data products and acquisition

The recommended route (matching the SMICA observed map) is the FFP10 component-separated simulation library:

- `dx12_v3_smica_cmb_mc_<00000..00999>_raw.fits` — nominally 1000 CMB (signal) MC, of which **999 are usable**: realization 00970 is a known missing/corrupt file on the PLA (ESA/PLA confirmation pending);
- `dx12_v3_smica_noise_mc_<00000..00299>_raw.fits` — 300 noise/systematics MC.

The E2E null is CMB+noise, paired *by parsed MC id*, `cmb[id] + noise[id mod  $n_{\text{noise}}$ ]` (the 300 noise MC cycled across the CMB set, matching the Planck-2018 permutation scheme). Id-based pairing is used precisely so the missing 00970 leaves a single absent realization rather than misaligning every later pairing. The ensemble is therefore the **PLA-available** set (999 usable CMB MC + 300 noise MC), not all 1000. At full IQU $N_{\text{side}} = 2048$ this is ~ 1 TB; a dedicated 2 TB nvme accommodates the full set. The observed maps and common temperature mask are already local. Raw maps stay outside version control; only compact summaries and provenance are committed. A per-simulation manifest records map IDs, URLs, SHA-256, beam, input N_{side} , analysis N_{side} , mask, and method.

5.3 Pipeline

The frozen pipeline is identical to the observed-map path: read each simulation (temperature field only), downgrade-on-read to the analysis N_{side} , apply the matched low- ℓ mask, compute the six registered statistics (`s_one_half`, `parity_even_over_odd_ratio`, `parity_asymmetry`, `planarity_mean`, `qo_axis_alignment_deg`, `axis_to_cmb_dipole_deg`), and feed the observed 6-vector and the simulation $N \times 6$ matrix to the look-elsewhere max-scan, which returns the +1-corrected global rank p -value (global $\geq$ min-local). Processing is streaming: peak memory is one map at a time, so disk and wall-time — not RAM — are the constraints.

5.4 Implemented runner modes

Three modes are implemented in `scripts/k1_global_maxscan.py`, each writing a *separate* artifact so the canonical GRF result is untouched (Table 2).

Table 2: K1 E2E runner modes. Only the full Route-A null is the exit gate that would re-point K1 from `measured_partial` to `measured`.

Mode	Null	Status / artifact
Route 4 (noise-only)	local Λ CDM signal + real per-method noise MC	<code>measured_partial</code> ; <code>*_e2e_noise.json</code>
Route A (full E2E)	real CMB MC + real noise MC (matched)	exit gate; <code>*_e2e_full.json</code>
v2 precision (either)	adds $\text{proc-}N_{\text{side}}$, ℓ_{max} , mask+inpaint	re-registered statistic set

5.5 Higher-precision (v2) statistic set

On a 12-core/64 GB workstation the analysis is I/O-bound, so precision can be raised at essentially the same wall-time. The v2 set (`htt/obsstat/lowell_precision.py`) is a deliberately *re-registered* statistic set (own configuration hash and `statistic.set` tag) with three levers:

1. **Processing N_{side} (default 64).** Raising N_{side} does *not* add information at $\ell \leq 8$ ($N_{\text{side}} = 16$ is Nyquist-sufficient); the only lever there is removing the pixel-window suppression (Table 3). $N_{\text{side}} = 64$ is the ceiling for $\ell \leq 8$; higher values help only if ℓ_{max} is raised.
2. **ℓ_{max} (default 30).** The ℓ -summed statistics ($S_{1/2}$, parity, planarity) gain genuine information from more multipoles; the quadrupole-octupole *alignment* statistics are intrinsically $\ell = 2, 3$ and are unchanged.
3. **Galactic mask + diffuse inpainting.** The common temperature mask is applied and the masked region diffuse-inpainted before the spherical-harmonic transform, *identically* on the observed map and every simulation, so the mask response is absorbed into the null

and the look-elsewhere p -value stays valid (no per-statistic deconvolution is claimed). The observed vector is taken from the full-resolution observed map, processed identically.

Table 3: Measured pixel-window w_ℓ (1.0 = no suppression). $N_{\text{side}} = 64$ removes the $N_{\text{side}} = 16$ suppression at $\ell = 6$ –8; higher N_{side} adds nothing at $\ell \leq 8$.

N_{side}	w_2	w_6	w_8
16	0.9989	0.9924	0.9870
64	0.9999	0.9995	0.9992
128	1.0000	0.9999	0.9998

5.6 Compute budget

Parallelism (`--jobs`) fans the per-simulation loop over processes (`spawn` context with one thread per worker, avoiding the fork-after-OpenMP deadlock); the noise set is pre-downgraded once and cached. The measured per-map cost at the v2 configuration is ≈ 3 s; the PLA-available 999+300 ensemble is I/O-bound (a ~ 3.5 min nvme read floor), giving the wall-times of Table 4. Peak memory is ≈ 1 GB per worker, so 12 workers use ≈ 12 GB and 24 use ≈ 24 GB — well within 64 GB. No additional RAM is required.

Table 4: Wall-time (PLA-available 999 CMB + 300 noise; noise pre-downgraded and cached).

Configuration	Serial	<code>--jobs</code> 12 (Ryzen 5900X)
v1 ($N_{\text{side}} 16, \ell \leq 8$)	~ 40 min	~ 5 –7 min
v2 ($N_{\text{side}} 64, \ell = 30$, mask)	~ 40 min	~ 5 –7 min
v2, <code>--max-sims</code> 300	~ 10 min	~ 2 min

5.7 Exit gate and kill switches

K1 re-points `measured_partial`→`measured` only through the ledger generator, with: a global rank p under the *matched* E2E ensemble; a configuration hash bound to {statistic list, mask, beam, ℓ -range, method}; an ensemble provenance manifest; and both component-separation methods reported side by side. The run stops if the map/mask/beam path differs between observed and simulations, if methods are mixed without labels, if fewer simulations are available than declared, if the statistic set is unfrozen, or if any text would claim E2E global significance without the matched provenance.

5.8 Co-evolution with the native low- ℓ solver

The v2 data analysis uses $\ell = 2$ –30. For internal consistency, the future native low- ℓ atlas should predict the *same* statistics over the same range: an atlas restricted to $\ell \leq 6$ cannot be scored against an $\ell = 2$ –30 morphology statistic, and Bianchi families (notably the VII_h spiral) diverge most at intermediate ℓ , where the discriminating power lives. The shear Fisher floor also keeps improving to $\ell \sim 20$ before saturating. The recommended atlas ceiling is therefore $\ell = 2$ –30 (staged: $\ell \leq 20$ for $\sim 90\%$ of the discrimination and floor gain, then $\ell \leq 30$ for full atlas–data parity), conditioned on the exact-FLRW-first gate and per- ℓ convergence. This remains separate (PR10) work and does not, by itself, grant any detection authority.

6 Claim envelope

The defensible envelope is a diagnostic methods-and-calibration framework: a rank-aware, provenance-bound sector comparator; conditional EGS-type theorems under stated hypotheses; and three model-independent real-data descriptors. It does *not* establish a globally significant low- ℓ anomaly, an anisotropic geometry, a Bianchi-family identification, a native-solver validation, or a physical-vorticity detection; no diagnostic is used as posterior odds and no scalar is promoted to a family. The Planck-raw E2E analysis narrows the K1 systematics gap; it does not change these boundaries.

A Reproducibility (commands)

Full Route-A E2E, v2 precision, parallel:

```
python scripts/k1_global_maxscan.py \  
  --cmb-mc-dir <cmb> --noise-mc-dir <noise> --method smica \  
  --max-sims 1000 --precision --proc-nside 64 --lmax 30 --jobs 12
```

Route-4 noise-only fallback:

```
python scripts/k1_global_maxscan.py --noise-mc-dir <noise> --method smica
```

Other entry points:

```
consolidated ledger: python scripts/build_egs_results_table.py  
download/run guide: docs/research_program/K1_E2E_DOWNLOAD_GUIDE.md  
standing blockers: docs/research_program/BLOCKERS.md
```